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Specifications



Market Data Production

APIs to access Market Data

Market Data

1.0.0 OAS3

Trader API - Market data

[Contact Schwab Trader API team](#)

Servers

<https://api.schwabapi.com/marketdata/v1>

Authorize



GET /quotes Get Quotes by list of symbols.



GET /{symbol_id}/quotes Get Quote by single symbol.



Option Chains Get Option Chains Web Service.



GET /chains Get option chain for an optionable Symbol



Option Expiration Chain Get Option Expiration Chain Web Service.



GET /expirationchain Get option expiration chain for an optionable symbol



PriceHistory Get Price History Web Service.



GET /pricehistory Get PriceHistory for a single symbol and date ranges.



Movers Get Movers Web Service.



GET /movers/{symbol_id} Get Movers for a specific index.



MarketHours Get MarketHours Web Service.



GET /markets Get Market Hours for different markets.



GET /markets/{market_id} Get Market Hours for a single market.



Instruments Get Instruments Web Service.





GET /instruments/{cusip_id} Get instrument by specific cusip

▼

Schemas



```
Bond  {
  cusip           [...]
  symbol          [...]
  description     [...]
  exchange        [...]
  assetType       [...]
  bondFactor      [...]
  bondMultiplier  [...]
  bondPrice       [...]
  type            [...]
}
```

high52	[...]
low52	[...]
dividendAmount	[...]
dividendYield	[...]
dividendDate	[...]
peRatio	[...]
pegRatio	[...]
pbRatio	[...]
prRatio	[...]
pcfRatio	[...]
grossMarginTTM	[...]
grossMarginMRQ	[...]
netProfitMarginTTM	[...]
netProfitMarginMRQ	[...]
operatingMarginTTM	[...]
operatingMarginMRQ	[...]
returnOnEquity	[...]
returnOnAssets	[...]
returnOnInvestment	[...]
quickRatio	[...]
currentRatio	[...]
interestCoverage	[...]
totalDebtToCapital	[...]
ltDebtToEquity	[...]
totalDebtToEquity	[...]
epsTTM	[...]
epsChangePercentTTM	[...]
epsChangeYear	[...]
epsChange	[...]
revChangeYear	[...]
revChangeTTM	[...]
revChangeIn	[...]
sharesOutstanding	[...]
marketCapFloat	[...]
marketCap	[...]

```

    divGrowthRate3Year      [...]
    dividendPayAmount       [...]
    dividendPayDate         [...]
    beta                    [...]
    vol1DayAvg               [...]
    vol10DayAvg              [...]
    vol3MonthAvg             [...]
    avg10DaysVolume          [...]
    avg1DayVolume            [...]
    avg3MonthVolume          [...]
    declarationDate          [...]
    dividendFreq              [...]
    eps                      [...]
    corpactionDate           [...]
    dtnVolume                [...]
    nextDividendPayDate      [...]
    nextDividendDate         [...]
    fundLeverageFactor        [...]
    fundStrategy              [...]
}

```

```

    symbol          [...]
    description     [...]
    exchange        [...]
    assetType       [...]
    type            [...]
}

```

```

InstrumentResponse {
    cusip           [...]
    symbol          [...]
    description     [...]
    exchange        [...]
    assetType       [...]
    bondFactor      [...]
    bondMultiplier  [...]
    bondPrice       [...]
    fundamental     FundamentalInst  {...}
    instrumentInfo   Instrument      {...}
    bondInstrumentInfo Bond           {...}
    type            [...]
}

```

```
    marketType      [...]
    exchange         [...]
    category         [...]
    product          [...]
    productName      [...]
    isOpen           [...]
    sessionHours     {...}
}
```

```
Interval {
    start      [...]
    end        [...]
}
```

```
Screener {
    description: Security info of most moved with in an index
    change      [...]
    description  [...]
    direction   [...]
    last        [...]
    symbol      [...]
    totalVolume [...]
}
```

```
    datetime            [...]
    datetimeISO8601     [...]
    high                [...]
    low                 [...]
    open                [...]
    volume              [...]
```

```
CandleList {
    candles            [...]
    empty              [...]
    previousClose      [...]
    previousCloseDate  [...]
    previousCloseDateISO8601 [...]
    symbol             [...]
```

```
EquityResponse {
    description:      Quote info of Equity security
    assetMainType     AssetMainType      [...]
    assetSubType      EquityAssetSubType  [...]
    ssid              [...]
    symbol             [...]
    realtime          [...]
    quoteType         QuoteType          [...]
    extended          ExtendedMarket    {...}
    fundamental       Fundamental       {...}
    quote             QuoteEquity       {...}
    reference         ReferenceEquity    {...}
    regular           RegularMarket     {...}
```

```

        invalidCusips          [...]
        invalidSSIDs           [...]
        invalidSymbols         [...]
    }

```

ExtendedMarket {

```

    description:      Quote data for extended hours

    askPrice          [...]
    askSize           [...]
    bidPrice          [...]
    bidSize           [...]
    lastPrice         [...]
    lastSize          [...]
    mark              [...]
    quoteTime         [...]
    totalVolume       [...]
    tradeTime         [...]
}

```

ForexResponse {

```

    description:      Quote info of Forex security

    assetMainType     AssetMainType  [...]
    ssid              [...]
    symbol             [...]
    realtime          [...]
    quote             QuoteForex    {...}
    reference          ReferenceForex {...}
}

```

```

    avg10DaysVolume      [...]
    avg1YearVolume       [...]
    declarationDate      [...]
    divAmount            [...]
    divExDate            [...]
    divFreq              DivFreq  [...]
    divPayAmount         [...]
    divPayDate           [...]
    divYield             [...]
    eps                  [...]
    fundLeverageFactor   [...]
    fundStrategy         FundStrategy  [...]
    nextDivExDate        [...]
    nextDivPayDate       [...]
    peRatio              [...]
}

```

```

FutureOptionResponse {
    description:      Quote info of Future Option security
    assetMainType     AssetMainType  [...]
    ssid             [...]
    symbol           [...]
    realtime         [...]
    quote            QuoteFutureOption  {...}
    reference        ReferenceFutureOption  {...}
}

```

```

    assetMainType      AssetMainType    [...]
    ssid               [...]
    symbol             [...]
    realtime           [...]
    quote              QuoteFuture      {...}
    reference           ReferenceFuture   {...}
}

```

```

IndexResponse {
    description:      Quote info of Index security
    assetMainType     AssetMainType    [...]
    ssid              [...]
    symbol            [...]
    realtime          [...]
    quote             QuoteIndex       {...}
    reference          ReferenceIndex   {...}
}

```

```

MutualFundResponse {
    description:      Quote info of MutualFund security
    assetMainType     AssetMainType    [...]
    assetSubType      MutualFundAssetSubType  [...]
    ssid              [...]
    symbol            [...]
    realtime          [...]
    fundamental       Fundamental      {...}
    quote             QuoteMutualFund   {...}
    reference          ReferenceMutualFund {...}
}

```



```
    assetMainType      AssetMainType    [...]
    ssid               [...]
    symbol              [...]
    realtime            [...]
    quote               QuoteOption     {...}
    reference           ReferenceOption  {...}
}
```

```

52WeekHigh          [...]
52WeekLow           [...]
askMICId            [...]
askPrice            [...]
askSize             [...]
askTime            [...]
bidMICId            [...]
bidPrice            [...]
bidSize            [...]
bidTime            [...]
closePrice          [...]
highPrice           [...]
lastMICId           [...]
lastPrice           [...]
lastSize           [...]
lowPrice            [...]
mark                [...]
markChange          [...]
markPercentChange   [...]
netChange           [...]
netPercentChange    [...]
openPrice           [...]
quoteTime           [...]
securityStatus      [...]
totalVolume         [...]
tradeTime           [...]
volatility          [...]
}

```

```

52WeekHigh          [...]
52WeekLow           [...]
askPrice             [...]
askSize              [...]
bidPrice             [...]
bidSize              [...]
closePrice           [...]
highPrice            [...]
lastPrice            [...]
lastSize             [...]
lowPrice             [...]
mark                 [...]
netChange            [...]
netPercentChange     [...]
openPrice            [...]
quoteTime            [...]
securityStatus       [...]
tick                 [...]
tickAmount           [...]
totalVolume          [...]
tradeTime            [...]
}

```

```

    askMICId          [...]
    askPrice           [...]
    askSize            [...]
    askTime            [...]
    bidMICId           [...]
    bidPrice           [...]
    bidSize            [...]
    bidTime            [...]
    closePrice         [...]
    futurePercentChange [...]
    highPrice          [...]
    lastMICId          [...]
    lastPrice          [...]
    lastSize           [...]
    lowPrice           [...]
    mark               [...]
    netChange          [...]
    openInterest       [...]
    openPrice          [...]
    quoteTime          [...]
    quotedInSession    [...]
    securityStatus     [...]
    settleTime         [...]
    tick               [...]
    tickAmount         [...]
    totalVolume        [...]
    tradeTime          [...]
}

```

```
askMICId          [...]
askPrice          [...]
askSize           [...]
bidMICId          [...]
bidPrice          [...]
bidSize           [...]
closePrice        [...]
highPrice         [...]
lastMICId         [...]
lastPrice         [...]
lastSize          [...]
lowPrice          [...]
mark              [...]
markChange        [...]
netChange         [...]
netPercentChange  [...]
openInterest      [...]
openPrice         [...]
quoteTime         [...]
securityStatus    [...]
settlemetPrice    [...]
tick              [...]
tickAmount        [...]
totalVolume       [...]
tradeTime         [...]
}
```



```
52WeekHigh      [...]
52WeekLow        [...]
closePrice       [...]
highPrice        [...]
lastPrice        [...]
lowPrice         [...]
netChange        [...]
netPercentChange [...]
openPrice        [...]
securityStatus   [...]
totalVolume      [...]
tradeTime        [...]
}
```

```
QuoteMutualFund {
  description:      Quote data of Mutual Fund security
  52WeekHigh        [...]
  52WeekLow          [...]
  closePrice         [...]
  nAV                [...]
  netChange          [...]
  netPercentChange   [...]
  securityStatus     [...]
  totalVolume        [...]
  tradeTime          [...]
}
```

52WeekHigh	[...]
52WeekLow	[...]
askPrice	[...]
askSize	[...]
bidPrice	[...]
bidSize	[...]
closePrice	[...]
delta	[...]
gamma	[...]
highPrice	[...]
indAskPrice	[...]
indBidPrice	[...]
indQuoteTime	[...]
impliedYield	[...]
lastPrice	[...]
lastSize	[...]
lowPrice	[...]
mark	[...]
markChange	[...]
markPercentChange	[...]
moneyIntrinsicValue	[...]
netChange	[...]
netPercentChange	[...]
openInterest	[...]
openPrice	[...]
quoteTime	[...]
rho	[...]
securityStatus	[...]
theoreticalOptionValue	[...]
theta	[...]
timeValue	[...]
totalVolume	[...]
tradeTime	[...]
underlyingPrice	[...]
vega	[...]

```

QuoteRequest {
  description:      Request one or more quote data in POST body

  cusips             [...]
  fields              [...]
  ssids              [...]
  symbols            [...]
  realtime           [...]
  indicative         [...]
}

```

```

QuoteResponse {
  description:      a (symbol, QuoteResponse) map. SCHW is an
                     example key

  < * >:             QuoteResponseObject  {...}
}

```

```

QuoteResponseObject {
  oneOf ->
    EquityResponse    {...}
    OptionResponse   {...}
    ForexResponse    {...}
    FutureResponse   {...}
    FutureOptionResponse {...}
    IndexResponse    {...}
    MutualFundResponse {...}
    QuoteError       {...}
}

```

```

    cusip                [...]
    description           [...]
    exchange             [...]
    exchangeName         [...]
    fsiDesc              [...]
    htbQuantity          [...]
    htbRate              [...]
    isHardToBorrow       [...]
    isShortable          [...]
    otcMarketTier        [...]
}

```

```

ReferenceForex {
    description:          Reference data of Forex security

    description           [...]
    exchange             [...]
    exchangeName         [...]
    isTradable           [...]
    marketMaker          [...]
    product              [...]
    tradingHours         [...]
}

```

```
    description          [...]
    exchange             [...]
    exchangeName         [...]
    futureActiveSymbol   [...]
    futureExpirationDate [...]
    futureIsActive       [...]
    futureMultiplier     [...]
    futurePriceFormat    [...]
    futureSettlementPrice [...]
    futureTradingHours   [...]
    product              [...]
}
```

```
ReferenceFutureOption {
    description:          Reference data of Future Option security
    contractType          ContractType  [...]
    description           [...]
    exchange              [...]
    exchangeName          [...]
    multiplier            [...]
    expirationDate        [...]
    expirationStyle       [...]
    strikePrice           [...]
    underlying            [...]
}
```



```
    description          [...]
    exchange             [...]
    exchangeName         [...]
  }
```

```
ReferenceMutualFund {
  description:      Reference data of MutualFund security

  cusip            [...]
  description      [...]
  exchange         [...]
  exchangeName     [...]
}
```

```
contractType      ContractType      [...]
cusip              [...]
daysToExpiration  [...]
deliverables       [...]
description        [...]
exchange           [...]
exchangeName       [...]
exerciseType       ExerciseType      [...]
expirationDay      [...]
expirationMonth    [...]
expirationType     ExpirationType    [...]
expirationYear     [...]
isPennyPilot       [...]
lastTradingDay     [...]
multiplier         [...]
settlementType     SettlementType    [...]
strikePrice        [...]
underlying         [...]
}
```

```
RegularMarket {
  description:          Market info of security

  regularMarketLastPrice      [...]
  regularMarketLastSize       [...]
  regularMarketNetChange      [...]
  regularMarketPercentChange  [...]
  regularMarketTradeTime      [...]
}
```

Enum:

Array [8]

EquityAssetSubType string

nullable: true

Asset Sub Type (only there if applicable)

Enum:

Array [11]

MutualFundAssetSubType string

nullable: true

Asset Sub Type (only there if applicable)

Enum:

Array [4]

ContractType string

Indicates call or put

Enum:

Array [2]

SettlementType string

option contract settlement type AM or PM

Enum:

Array [2]

business day of the month), Q for Quarterly expirations (last business day of the quarter month MAR/JUN/SEP/DEC), W for Weekly expiration (also called Friday Short Term Expirations) and S for Expires 3rd Friday of the month (also known as regular options).

Enum:

Array [4]

FundStrategy string

nullable: true

FundStrategy "A" - Active "L" - Leveraged "P" - Passive "Q" - Quantitative "S" - Short

Enum:

Array [6]

ExerciseType string

option contract exercise type America or European

Enum:

Array [2]

DivFreq integer

nullable: true

Dividend frequency 1 - once a year or annually 2 - 2x a year or semi-annually 3 - 3x a year (ex. ARCO, EBRPF) 4 - 4x a year or quarterly 6 - 6x per yr or every other month 11 - 11x a year (ex. FBND, FCOR) 12 - 12x a year or monthly

Enum:

Array [8]

NBBO - realtime, NFL - Non-fee liable quote.

Enum:

Array [3]

```
ErrorResponse {
  errors          [...]
}
```

```
Error {
  id              [...]
  status          [...]
  title           [...]
  detail          [...]
  source          ErrorSource {...}
}
```

```
ErrorSource {
  description:    Who is responsible for triggering these errors.
  pointer         [...]
  parameter       [...]
  header          [...]
}
```



```
status [...]
underlying Underlying {...}
strategy [...]
interval [...]
isDelayed [...]
isIndex [...]
daysToExpiration [...]
interestRate [...]
underlyingPrice [...]
volatility [...]
callExpDateMap {...}
putExpDateMap {...}
}
```

```
OptionContractMap {
  < * >: OptionContract {...}
}
```

```

askSize          [...]
bid              [...]
bidSize         [...]
change          [...]
close           [...]
delayed         [...]
description     [...]
exchangeName    [...]
fiftyTwoWeekHigh [...]
fiftyTwoWeekLow [...]
highPrice       [...]
last            [...]
lowPrice        [...]
mark            [...]
markChange      [...]
markPercentChange [...]
openPrice       [...]
percentChange    [...]
quoteTime       [...]
symbol          [...]
totalVolume     [...]
tradeTime       [...]
}

```

```

OptionDeliverables {
    symbol          [...]
    assetType       [...]
    deliverableUnits [...]
    currencyType    [...]
}

```

symbol	[...]
description	[...]
exchangeName	[...]
bidPrice	[...]
askPrice	[...]
lastPrice	[...]
markPrice	[...]
bidSize	[...]
askSize	[...]
lastSize	[...]
highPrice	[...]
lowPrice	[...]
openPrice	[...]
closePrice	[...]
totalVolume	[...]
tradeDate	[...]
quoteTimeInLong	[...]
tradeTimeInLong	[...]
netChange	[...]
volatility	[...]
delta	[...]
gamma	[...]
theta	[...]
vega	[...]
rho	[...]
timeValue	[...]
openInterest	[...]
isInTheMoney	[...]
theoreticalOptionValue	[...]
theoreticalVolatility	[...]
isMini	[...]
isNonStandard	[...]
optionDeliverablesList	[...]
strikePrice	[...]
expirationDate	[...]

```

    multiplier                [...]
    settlementType             SettlementType  [...]
    deliverableNote            [...]
    isIndexOption              [...]
    percentChange              [...]
    markChange                 [...]
    markPercentChange          [...]
    isPennyPilot               [...]
    intrinsicValue             [...]
    optionRoot                 [...]
}

```

```

ExpirationChain {
    status                [...]
    expirationList        [...]
}

```

```

Expiration {
    description:           expiration type
    daysToExpiration       [...]
    expiration             [...]
    expirationType         ExpirationType  [...]
    standard               [...]
    settlementType         SettlementType   [...]
    optionRoots            [...]
}

```



Developer Portal